LESSON PLAN

EJECTION SEAT (36)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	01
Total No. of Lesson Plans	6	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Difference between conventional and nuclear explosions
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Difference between conventional and nuclear explosions
5. Specific Objectives	36.1.1.1 Trainee will be able to explain conventional and nuclear explosion

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of theoretical differences; No hands-on practical activity
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary source of energy in a conventional explosion?
(b) Define the term 'chain reaction' in the context of nuclear physics.
(c) List two key differences between chemical and nuclear energy release.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Conventional Explosion Mechanism • Chemical Bond Energy Release • Shock Wave Characteristics • Yield Measurement Units • Environmental Impact Factors	L & Ds	PPT, IB	15 min
(b)	• Nuclear Explosion Mechanism • Fission vs Fusion Processes • Radioactive Fallout Effects • Blast Radius Comparison • Thermal Radiation Intensity	L & Ds	PPT, PDF	20 min
(c)	• Strategic Implications of Yield • Detection and Monitoring Systems • Safety Protocols for Personnel • Historical Case Studies • Future Technological Trends	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain why nuclear explosions produce significantly more thermal radiation than conventional ones.
(b) Describe the role of radioactive fallout in distinguishing nuclear from conventional blasts.
(c) Compare the speed of energy release in fission versus chemical reactions.

12. Home Assignment and Follow Up	Prepare a brief note on how the atomic structure concepts covered in LP 02 relate to the mechanism of nuclear fission discussed in this lesson.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Definitions of Atomic structure, Nucleus Protons, Neutrons Electrons Atomic mass Isotopes Mass defect Nuclear fission Nuclear fusion Fission –fusion –fission	Refer to the master syllabus reference list	LP 02

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min